Government of Karnataka

DEPARTMENT OF SCHOOL EDUCATION

(PRE-UNIVERSITY)

SECOND YEAR P U C

COMPUTER SCIENCE (NEW SYLLABUS)

PRACTICAL MANUAL

2025 - 26

List of practical problems - II PU Course

PART – A

Python

1) Write a python program using a function to print n fibonacci numbers.
2) Write a menu driven program in python to find factorial and sum of natural of n Numbers using function.
3) Write a python program using user defined function to calculate interest amount using simple interest method and compound interest method and find the difference of interest amount between the two methods.
4) Write a Python Program to create a text file and to read a text file and display the number of vowels, consonants, uppercase and lowercase characters in the file.
5) Write a python code to find the number of lines, number of words and number of characters in a text file.
6) Write a python program to create and to read records in binary file with student name and marks of six subjects.
7) Write a python program to copy the records of the students having percentage 90 and above from the binary file into another file.
8) Write a python program using function to sort the elements of a list using bubble sort method.
9) Write a python program using function to sort the elements of a list using selection sort method.
10) Write a python program using function to sort the elements of a list using insertion sort method.
11) Write a python program using function to search an element in a list using linear search method.
12) Write a python program using function to search an element in a list using binary search method.
13) Write a python program to add and display elements from a stack using list.
14) Write a python program to add and display elements from a queue using list.

DSE (PUE)

PART B

MYSQL

B1) Create a table with the following fields and enter 10 records into the table.
Entity Name: marks

Attribute name	Type	Size	Constraints
Rollno	Int	5	
Sname	Varchar	15	Not null
Lang_mks	Int	3	Between 0 and 100
Eng_mks	Int	3	Between 0 and 100
Sub1_mks	Int	3	Between 0 and 100
Sub2_mks	Int	3	Between 0 and 100
Sub3_mks	Int	3	Between 0 and 100
Sub4_mks	Int	3	Between 0 and 100

(i) List all the records
(ii) Display the description of the table
(iii) Add the new attributes total and percent
(iv) Calculate total and percentage of marks for all the students
(v) List the students whose percentage of marks is more than 60%.
(vi) List the students whose percentage is between 60 % and 85%.
(vii) Arrange the students based on percentage of marks from highest to lowest.

B2) Create a table for house hold Electricity bill with the following fields and enter 10 records.
Entity Name: BESCOM

Attributename	Type	Size	Constraint
RRNO	Varchar	10	Primary key
CUSTNAME	Varchar	25	Not null
BILLDATE	DATE		
UNITS	INT	4	

1. View the structure of table.
2. List all the records
3. Add a new field for bill amount in the name of billamt.
4. Compute the bill amount for each consumer as per the following rules.
   a. MINIMUM Amount Rs. 100
   b. For first 100 units Rs 7.50/Unit
   c. For the above 100 units Rs. 8.50/Unit
5. Display the maximum, minimum and total bill amount.
6. List all the bills generated in a sorted order based on RRNO.

B3) Create a table with the following details and enter 10 records into the table.

Entity Name: student

Attribute name	Type	Size	Constraint
Rollno	int	5	Primary key
Sname	Varchar	15	Not null
DOB	date		
Gender	char	1	
Combn	Varchar	5	
Class	Char	6	

(i) List all the students
(ii) List the students who are in BASC and CEBA combination.
(iii) List only the combination by removing duplicate values.
(iv) List the students alphabetically.
(v) List the students alphabetically and class-wise.
(vi) List the students who born in the month of June of any year.
(vii) Count the number of students Gender-wise.

DSE (PUE)

B4) Create a table with following fields and enter 10 records into the table.

Entity Name: Library

Attributename	**Type**	**Size**	**Constraint**
Title	Varchar	75	Not null
Author	Varchar	60	
Year	int	4	
Category	Varchar	25	
Price	float	7,2	
Qty	Int	4	

(i) List all the books
(ii) Calculate Amount by altering table with a new column 'Amount'
(iii) List the records of all those books price is between 400 and 900.
(iv) List those records with no value in the attribute year .
(v) List the names of the authors whose name starts with 'C' or 'D'.
(vi) List Title, year, category with category has word 'science'.
(vii) List all those records whose year of publication is 2010 onwards with price is less than 750.

DSE (PUE)

PRACTICAL EXAMINATION

Sl. No	Program	Marks
1	Writing 1 Program from PART-A (Python)	6 Marks
2	Writing 1 Program from PART-B (SQL)	6 Marks
3	Execution of any one program of examiner's choice	8 Marks
4	Practical record	6 Marks
5	Viva-voce (Four Questions related to the program written by the student in Exam)	4 Marks
Total		30 Marks

PRACTICAL MARKS DISTRIBUTION

PART-A	Writing One Python program	06
PART-B	Writing One SQL Program (Commands)	06
Execution of <u>any one program</u> of examiner's choice		08
Viva-Voce (2-Internal, 2-External) (Questions related to the program written by the student in Practical Exam)		04
Practical Record (Marks)	Attendance	
	90% - 100%	06
	80% - 90 %	05
	70% - 80%	04
	60% - 70%	03
	Below 60%	00
TOTAL →		30

DSE (PUE)

PART A

PYTHON

A1. Write a python program using a function to print fibonacci series up to n numbers

```
# Fibonacci numbers
def fibo(n):
    num1 = 0
    num2 = 1
    next_number = num2
    count = 2
    print(num1," ", num2, end=" ")
    while count <n:
        print(next_number, end=" ")
        count += 1
        num1, num2 = num2, next_number
        next_number = num1 + num2
n = int(input("enter the Limit"))
fibo(n)
print()
```

OUTPUT:

enter the Limit 10

0 1 1 2 3 5 8 13 21 34 55 89

DSE (PUE)

A2. Write a Menu driven program in python to find factorial, and sum of natural Numbers using a function

```
# Python program to find factorial of given number and Sum of natural numbers
def fact(n):
   return 1 if (n==1 or n==0) else n * fact(n - 1);
def sum(n):
   return 0 if (n==0) else n+sum(n-1);
num = int(input("Enter any number : "))

print("1-To find the factorial, 2-To find the sum  3-Exit")
opt=int(input("Enter the option 1-3 : "))
if (opt==1):
   print("Factorial of ",num,"is : ",fact(num))
else:
      if(opt==2):
        print("Sum of ",num,"is : ",sum(num))
      else:
         print(" ")
```

OUTPUT:

Enter any number : 5
1-To find the factorial, 2-To find the sum 3-Exit
Enter the option 1-3 : 1
Factorial of 5 is : 120

Enter any number : 5
1-To find the factorial, 2-To find the sum 3-Exit
Enter the option 1-3 : 2
Sum of 5 is : 15

DSE (PUE)

A3. Write a python program using user defined function to calculate interest amount using simple interest method and compound interest method and find the difference of interest amount between the two methods.

```
# Simple and Compound Interest
# Reading principal amount, rate and time
def simpInt(principle,time,rate):
   si = float(principle*time*rate/100)
   return si
def compInt(principle,time,rate):
   ci = float(principle * ((1+rate/100)**time - 1))
   return ci
principle = float(input('Enter amount: '))
time = float(input('Enter time: '))
rate = float(input('Enter rate: '))
si=simpInt(principle,time,rate)
ci=compInt(principle,time,rate)
print('Simple interest is Rs. %8.2f' % si)
print('Compound interest is Rs. %8.2f' % ci)
diffint=ci-si;
print("Difference is Rs. %8.2f " % diffint)
```

OUTPUT:

Enter amount: 45000
Enter time: 3
Enter rate: 12.5
Simple interest is Rs. 16875.00
Compound interest is Rs. 19072.27
Difference is Rs. 2197.27

DSE (PUE)

A4. Write a Python Program to read a text file and display the number of vowels, consonants, uppercase and lowercase characters in the file

```
# Python Program to create a text file and to read a text file and display the number of
# vowels, consonants, uppercase and lowercase characters in the file
def count_characters(file_name):
    vowels = "aeiouAEIOU"
    vowel_count = 0
    consonant_count = 0
    uppercase_count = 0
    lowercase_count = 0

    with open(file_name, "r") as file:
        text = file.read()

    for char in text:
        if char.isalpha():
            if char in vowels:
                vowel_count += 1
            else:
                consonant_count += 1

            if char.isupper():
                uppercase_count += 1
            elif char.islower():
                lowercase_count += 1

    print("Vowels:" , vowel_count)
    print("Consonants: ", consonant_count)
    print("Uppercase characters: ",uppercase_count)
    print("Lowercase characters: ", lowercase_count)

def create_text_file(file_name, content):
    with open(file_name, "w") as file:
        file.write(content)

file_name = "sample.txt"
content = input("enter few sentences to create a text file with content: ")
create_text_file(file_name, content)

count_characters(file_name)
```

OUTPUT:

enter few sentences to create a text file with content: This program is TO Create a FILE
Vowels: 11
Consonants: 15
Uppercase characters: 8
Lowercase characters: 18

DSE (PUE)

A5) Write a python code to count the number of lines, number of words and number of characters in a text file.

```
# Python program to count number of lines,
# Words and character in textfile
def count_text_file(file_name):
    line_count = 0
    word_count = 0
    char_count = 0

    with open(file_name, "r") as file:
        for line in file:
            line_count += 1
            word_count += len(line.split())
            char_count += len(line)

    print("Lines: ", line_count)
    print("Words: ", word_count)
    print("Characters: ", char_count)


def create_text_file(file_name, content):
    with open(file_name, "w") as file:
        file.write(content)

file_name = "sample.txt"
content = """Hello Students
This is a sample text file.
It contains multiple lines."""
create_text_file(file_name, content)

count_text_file(file_name)
```

OUTPUT:

Lines: 3
Words: 12
Characters: 70

DSE (PUE)

A6) Write a python program to create and to read records in binary file with student name and marks of six subjects.

```
import pickle
while True:
    print('''    1. Create Binary File.
    2. Display the File.
    3. Exit''')
    a=int(input('choose a command (1-2,3-exit): '))
    if a==1:
        f=open('student.dat','wb')
        x=int(input('How many student: '))
        for i in range(x):
            name=input('Name: ')
            english=int(input('English Mark: '))
            lan=int(input("Language Marks"))
            phy=int(input('Physics Mark: '))
            chem=int(input('Chemistry Mark: '))
            maths=int(input('Maths Mark: '))
            cs=int(input('CS Mark: '))
            t=[name,english,lan,phy,chem,maths,cs]
            pickle.dump(t,f)
        f.close()
    elif a==2:
        f=open('student.dat','rb')
        try:
            while True:
                p=pickle.load(f)
                print(p)
        except:
          f.close()
    if a>2:
        break
```

OUTPUT:

1. Create Binary File.
2. Display the File.
3. Exit
choose a command (1-2, 3-exit): 2
['Bharath', 77, 88, 99, 77, 88, 99]
['Guru', 77, 88, 99, 77, 88, 99]
['Manjula', 88, 99, 88, 99, 88, 88]
['Rekha', 99, 88, 99, 99, 88, 88]
['Uday', 66, 77, 66, 77, 66, 77]
1. Create Binary File.
2. Display the File.
3. Exit
choose a command (1-2,9-exit):3

DSE (PUE)

A7. Write a python program to copy the records of the students having percentage 90 and above from the binary file into another file.

```
# program to create and copy records to copy records with percentage 90 and above into another file.
import pickle
while True:
    print('''    1. Create Binary File.
    2. Display the main File.
    3. Create new file wirh >90
    4. Exit''')
    a=int(input('choose a command (1-2,9-exit): '))
    if a==1:
        f=open('student.dat','wb')
        o=open('stuii.dat','wb')
        x=int(input('How many student: '))
        for i in range(x):
            name=input('Name: ')
            english=int(input('English Mark: '))
            lan=int(input("Language Marks"))
            phy=int(input('Physics Mark: '))
            chem=int(input('Chemistry Mark: '))
            maths=int(input('Maths Mark: '))
            cs=int(input('CS Mark: '))
            total=phy+chem+cs+maths+english+lan
            per=(total / 600) * 100
            t=[name,english,lan,phy,chem,maths,cs,total,per]
            g=[name,english,lan,phy,chem,maths,cs,total,per]
            pickle.dump(t,f)
            if per>=90:
              pickle.dump(g,o)
        f.close()
        o.close()
    elif a==2:
        f=open('student.dat','rb')
        try:
            while True:
                p=pickle.load(f)
                print(p)
        except:
          f.close()
    elif a==3:
        print("Studets with > 90 Marks")
        f=open('stuii.dat','rb')
        try:
            while True:
                o=pickle.load(f)
                print(o)
        except:
            f.close()
```

DSE (PUE)

```
    else:
        break
```

OUTPUT:

```
  1. Create Binary File.
  2. Display the main File.
  3. Create new file wirh >90
  4. Exit
choose a command (1-2,9-exit): 2
['Bharath', 77, 88, 99, 88, 77, 77, 506, 84.33333333333334]
['Guru', 88, 77, 88, 77, 88, 77, 495, 82.5]
['Manjula', 88, 77, 88, 88, 77, 77, 495, 82.5]
['Rekha', 99, 88, 99, 88, 99, 88, 561, 93.5]
['Uday', 99, 99, 99, 99, 99, 99, 594, 99.0]
  1. Create Binary File.
  2. Display the main File.
  3. Create new file wirh >90
  4. Exit
choose a command (1-2,9-exit): 3
Studets with > 90 Marks
['Rekha', 99, 88, 99, 88, 99, 88, 561, 93.5]
['Uday', 99, 99, 99, 99, 99, 99, 594, 99.0]
  1. Create Binary File.
  2. Display the main File.
  3. Create new file wirh >90
  4. Exit
choose a command (1-2,9-exit):4
```

DSE (PUE)

A8. Write a python program using function to sort the elements of a list using bubble sort method

```
# program using function to sort the elements of list using bubble sort method
def bubble_sort(arr):
    n = len(arr)
    for i in range(n):
        for j in range(0, n-i-1):
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]

def input_list():
    arr = []
    n = int(input("Enter the number of elements in the list: "))
    for i in range(n):
        element = int(input("Enter element "))
        arr.append(element)
    return arr

arr = input_list()
print("Original list:", arr)

bubble_sort(arr)
print("Sorted list:", arr)
```

OUTPUT:

Enter the number of elements in the list: 5
Enter element 34
Enter element 56
Enter element 23
Enter element 12
Enter element 34
Original list: [34, 56, 23, 12, 34]
Sorted list: [12, 23, 34, 34, 56]

DSE (PUE)

A9. Write a python program using function to sort the elements of a list using selection sort method

```
# Selection sort in Python
# Selection sort in Python
def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_index = i
        for j in range(i+1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        arr[i], arr[min_index] = arr[min_index], arr[i]

def input_list():
    arr = []
    n = int(input("Enter the number of elements in the list: "))
    for i in range(n):
        element = int(input("Enter element "))
        arr.append(element)
    return arr

arr = input_list()

print("Original list:", arr)

selection_sort(arr)
print("Sorted list:", arr)
```

OUTPUT:

Enter the number of elements in the list: 5
Enter element 32
Enter element 45
Enter element 67
Enter element 34
Enter element 23
Original list: [32, 45, 67, 34, 23]
Sorted list using insertion sort method: [23, 32, 34, 45, 67]

DSE (PUE)

A10. Write a python program using function to sort the elements of a list using insertion sort method

```
# python program using function to sort the elements of list using insertion sort method
def insertion_sort(arr):
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j >= 0 and key < arr[j]:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key

def input_list():
    arr = []
    n = int(input("Enter the number of elements in the list: "))
    for i in range(n):
        element = int(input("Enter element "))
        arr.append(element)
    return arr

arr = input_list()
print("Original list:", arr)

insertion_sort(arr)
print("Sorted list using insertion sort method:", arr)
```

OUTPUT:

Enter the number of elements in the list: 5
Enter element 55
Enter element 43
Enter element 23
Enter element 43
Enter element 65
Original list: [55, 43, 23, 43, 65]
Sorted list using insertion sort method: [23, 43, 43, 55, 65]

DSE (PUE)

A11. Write a python program using function to search an element in a list using linear search method

```
# Search function with parameter list name
# and the value to be searched - Linear Search

def linear_search(arr, target):
    for index, element in enumerate(arr):
        if element == target:
            return index
    return -1
def input_list():
    arr = []
    n = int(input("Enter the number of elements in the list: "))
    for i in range(n):
        element = int(input("Enter element "))
        arr.append(element)
    return arr

arr = input_list()
target = int(input("Enter the element to search for: "))

result = linear_search(arr, target)
if result != -1:
    print(target, " Element found at index ", result)
else:
    print(target, " Element not found in the list.")
```

OUTPUT:

Enter the number of elements in the list: 5
Enter element 23
Enter element 4
Enter element 5
Enter element 6
Enter element 45
Enter the element to search for: 55
55 Element not found in the list.

DSE (PUE)

A12. Write a python program using function to search an element in a list using binary search method.

```
def binary_search(arr, target):
low, high = 0, len(arr) - 1
    while low<= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1

def input_list():
    arr = []
    n = int(input("Enter the number of elements in the list: "))
    for i in range(n):
        element = int(input("Enter the elements in ascending order "))
        arr.append(element)
    return arr

arr = input_list()
target = int(input("Enter the element to search for: "))

result = binary_search(arr, target)
if result != -1:
    print(target, "Element found at index ", result)
else:
    print(target, "Element not found in the list.")
```

OUTPUT:

Enter the number of elements in the list: 5
Enter the elements in ascending order 12
Enter the elements in ascending order 23
Enter the elements in ascending order 34
Enter the elements in ascending order 454
Enter the elements in ascending order 656
Enter the element to search for: 34
34 Element found at index 2

DSE (PUE)

A13. Write a python program to add and display elements from a stack using list

```
# initial empty stack
stack = []
#PUSHing the elements into a stack
print("initially stack is empty :",stack)

stack.append('x')
stack.append('y')
stack.append('z')

print("After PUSHING stack is :")

print(stack)

#POPing the elements from a stack
print('After POPped from stack: ')
print(stack.pop())
print(stack.pop())
print(stack.pop())

print('\nmy_stack after elements are poped:')
print(stack)
```

OUTPUT:

initially stack is empty : []
After PUSHING stack is :
['x', 'y', 'z']
After POPped from stack:
z
y
x

my_stack after elements are poped:
[]

DSE (PUE)

A14. Write a python program to add and display elements from a queue using list

```
# python program to add and display elements from a queue using list
# Create Queue and perform insert and delete

import queue

# Display the elements in the queue
def display_queue(q):
    print("Queue elements are:", end=" ")

    while not q.empty():
        element = q.get()
        print(element, end=" ")
    print('\n Queue size after REMOVE is ', q.qsize())


# Create a Queue object
q = queue.Queue()

# Add elements to the queue using put()
q.put(10)
q.put(20)
q.put(30)
print('Queue size after INSERT is ', q.qsize())

# Display the elements in the queue
display_queue(q)
```

OUTPUT:

Queue size after INSERT is 3
Queue elements are: 10 20 30
Queue size after REMOVE is 0

DSE (PUE)

PART B

MYSQL

B1) Create a table with the following fields and enter 10 records into the table.
Entity Name: marks

Attribute name	Type	Size	Constraints
Rollno	Int	5	
Sname	Varchar	15	Not null
Lang_mks	Int	3	Between 0 and 100
Eng_mks	Int	3	Between 0 and 100
Sub1_mks	Int	3	Between 0 and 100
Sub2_mks	Int	3	Between 0 and 100
Sub3_mks	Int	3	Between 0 and 100
Sub4_mks	Int	3	Between 0 and 100

```
CREATE TABLE marks (
   Rollno INT(5),
   Sname VARCHAR(15) NOT NULL,
   Lang_mks INT(3) CHECK (Lang_mks BETWEEN 0 AND 100),
   Eng_mks INT(3) CHECK (Eng_mks BETWEEN 0 AND 100),
   Sub1_mks INT(3) CHECK (Sub1_mks BETWEEN 0 AND 100),
   Sub2_mks INT(3) CHECK (Sub2_mks BETWEEN 0 AND 100),
   Sub3_mks INT(3) CHECK (Sub3_mks BETWEEN 0 AND 100),
   Sub4_mks INT(3) CHECK (Sub4_mks BETWEEN 0 AND 100)
);
```

Data to be entered: (Values are indicative)

Rollno	Sname	Lang_mks	Eng_mks	Sub1_mks	Sub2_mks	Sub3_mks	Sub4_mks
1010	RAJ	89	97	98	99	86	95
1026	KIRAN	67	62	72	86	72	62
1042	ANAND	78	87	92	82	72	76
1250	RAM	72	86	72	62	87	68
5212	VIJAYA	46	58	86	92	72	62
3622	MANOJ	86	56	62	86	52	64
1948	REEHAN	63	68	52	56	96	76
1482	KAJOL	49	54	48	76	62	55
1947	KUMAR	98	98	99	100	97	99
1951	REEMA	82	72	62	98	73	64

```
INSERT INTO marks  VALUES(1010, 'RAJ', 89, 97, 98, 99, 86, 95)
INSERT INTO marks  VALUES (1026, 'KIRAN', 67, 62, 72, 86, 72, 62)
INSERT INTO marks  VALUES (1042, 'ANAND', 78, 87, 92, 82, 72, 76)
INSERT INTO marks  VALUES (1250, 'RAM', 72, 86, 72, 62, 87, 68)
INSERT INTO marks  VALUES (5212, 'VIJAYA', 46, 58, 86, 92, 72, 62)
INSERT INTO marks  VALUES (3622, 'MANOJ', 86, 56, 62, 86, 52, 64)
INSERT INTO marks  VALUES (1948, 'REEHAN', 63, 68, 52, 56, 96, 76)
INSERT INTO marks  VALUES (1482, 'KAJOL', 49, 54, 48, 76, 62, 55)
INSERT INTO marks  VALUES (1947, 'KUMAR', 98, 98, 99, 100, 97, 99)
```

DSE (PUE)

(i) List all the records

select * from marks;

(ii) Display the description of the table

describe marks;

(iii) Add the new attributes total and percent

alter table marks add(total int(3), percent float(5,3));

(iv) Calculate total and percentage of marks for all the students

update marks set total = lang_mks+eng_mks+sub1_mks+sub2_mks+sub3_mks+sub4_mks;

update marks set percent = total/600*100;

(v) List the students whose percentage of marks is more than 60%.

select sname, percent from marks where percent >=60;

(vi) List the students whose percentage is between 60% and 85%.

select sname, percent from marks where percent between 60 and 85;

(vii) Arrange the students based on percentage of marks from highest to lowest.

select sname, percent from marks order by percent desc;

DSE (PUE)

B2) Create a table for house hold Electricity bill with the following fields and enter 10 records.
Entity Name: BESCOM

Attributename	Type	Size	Constraint
RRNO	Varchar	10	Primary key
CUSTNAME	Varchar	25	Not null
BILLDATE	DATE		
UNITS	INT	4	

```
CREATE TABLE BESCOM (
   RRNO VARCHAR(10) PRIMARY KEY,
   CUSTNAME VARCHAR(25) NOT NULL,
   BILLDATE DATE,
   UNITS INT(4)
);
```

Data to be entered: (Values are indicative)

RRNO	CUSTNAME	BILLDATE	UNITS
E1120	RAJ	2024-05-5	250
E2210	KIRAN	2024-03-26	178
E1450	ANAND	2024-04-15	56
E2126	RAM	2024-05-8	782
E1562	MANJULA	2024-05-2	562
E6221	MANOJ	2024-05-18	72
E5822	REEHAN	2024-02-19	92
E1692	KAJOL	2024-03-25	73
E6721	KUMAR	2024-07-14	589
E2682	REEMA	2024-05-11	100

```
INSERT INTO BESCOM  VALUES('E1120', 'RAJ', '2024-05-05', 250)
INSERT INTO BESCOM  VALUES ('E2210', 'KIRAN', '2024-03-26', 178)
INSERT INTO BESCOM  VALUES ('E1450', 'ANAND', '2024-04-15', 56)
INSERT INTO BESCOM  VALUES ('E2126', 'RAM', '2024-05-08', 782)
INSERT INTO BESCOM  VALUES ('E1562', 'MANJULA', '2024-05-02', 562)
INSERT INTO BESCOM  VALUES ('E6221', 'MANOJ', '2024-05-18', 72)
INSERT INTO BESCOM  VALUES ('E5822', 'REEHAN', '2024-02-19', 92)
INSERT INTO BESCOM  VALUES ('E1692', 'KAJOL', '2024-03-25', 73)
INSERT INTO BESCOM  VALUES ('E6721', 'KUMAR', '2024-07-14', 589)
INSERT INTO BESCOM  VALUES ('E2682', 'REEMA', '2024-05-11', 100);
```

7. View the structure of table.
   describe bescom;

8. List all the records
   Select * from bescom;

9. Add a new field for bill amount in the name of billamt.
   ALTER TABLE BESCOM ADD BILLAMT FLOAT(7,2);

10. Compute the bill amount for each consumer as per the following rules.
    a. MINIMUM Amount Rs. 100
    b. For first 100 units Rs 7.50/Unit
    c. For the above 100 units Rs. 8.50/Unit

(i) UPDATE BESCOM SET BILLAMT = 100 + UNITS * 7.50 WHERE UNITS <= 100;

(ii) UPDATE BESCOM SET BILLAMT = 100 + (100 * 7.50) + (UNITS - 100) * 8.50 WHERE UNITS > 100;

11. Display the maximum, minimum and total bill amount.

SELECT MAX(BILLAMT), min(billamt), avg(billamt), sum(billamt) FROM BESCOM;

12. List all the bills generated in a sorted order based on RRNO.

    select rrno from bescom order by rrno;

DSE (PUE)

B3) Create a table with the following details and enter 10 records into the table.

Entity Name: student

Attribute name	Type	Size	Constraint
Rollno	int	5	Primary key
Sname	Varchar	15	Not null
DOB	date		
Gender	char	1	
Combn	Varchar	5	
Class	Char	6	

```
CREATE TABLE student (
   Rollno int(5) PRIMARY KEY,
   Sname VARCHAR(15) NOT NULL,
   dob date,
   Gender CHAR(1),
   Combn VARCHAR(5),
   Class varCHAR(6)
);
```

Data to be entered: (Values are indicative)

RollNo	Sname	DOB	Gender	Combn	Class

```
insert into student values (3739, 'Uday', '2004-9-12', 'M', 'PCMC','2B');
insert into student values (1001, 'Raj Kumar', '2005-5-21', 'M', 'BASC','2A');
insert into student values (1005, 'Kiran', '2004-11-15', 'M', 'PCMC','2B');
insert into student values (1042, ‘Anand', '2005-12-22', 'M', ‘CEBA','2C');
insert into student values (1250, 'Ram', '2004-6-18', 'M', 'PCMC','2A');
insert into student values (5212, 'Vijaya', '2007-7-28', 'F', 'PCMC','2A');
insert into student values (1029, 'Bharath', '2005-1-12', 'M', 'BASC','2B');
insert into student values (2152, 'Rekha', '2006-6-8', 'F', 'CEBA','2C');
insert into student values (1948, 'Reehan', '2005-4-17', 'M', 'CEBA','2C');
insert into student values (2443, 'Manjula', '2005-8-15', 'F', 'PCMC','2B');
```

(i) List all the students

select * from student;

(ii) List only those students who are in BASC and CEBA combination.

select * from student where combn='BASC' or combn='CEBA';

(iii) List only the combination by removing duplicate values.

select distinct(combn) from student;

(iv) List the students alphabetically.

select sname from student order by sname;

(v) List the students alphabeticallyclass-wise.

select sname, class from student group by class order by sname;

(vi) List the students who born in the month of June of any year.

SELECT * FROM student WHERE MONTH(dob) = 6;

(vii) Count the number of students Gender-wise.

SELECT gender, COUNT(*) FROM student GROUP BY gender;

DOE (PUE)

B4) Create a table with following fields and enter 10 records into the table.

Entity Name: Library

Attributename	Type	Size	Constraint
Title	Varchar	75	Not null
Author	Varchar	60	
Year	int	4	
Category	Varchar	25	
price	float	7,2	
Qty	Int	4	

```
CREATE TABLE Library (
   Title VARCHAR(75) NOT NULL,
   Author VARCHAR(60),
   Year INT(4),
   Category VARCHAR(25),
   Price FLOAT(7,2),
   Qty INT(4)
);
```

Data to be entered: (Values are indicative)

Title	Author	Year	Category	Price	Qty
The Data Science Handbook	Darshan	2019	Data science	750	12
Introduction to Computer Programs	Harshavardhan	null	Computer Science	700	6
Computer Science Text book Class 12	Reeta Sahoo	2019	Textbook	450	4
A book on AI	Sagar	2016	AI	200	7
Robots and Automation	Dushyanth	2018	Digital Technology	750	9
AI 2041	Chen & Lee	2021	AI	1000	2
Computer Hardware and Software	Chethan	2000	Computer science	500	7
The Future of Work	Dev Kumar	2018	Digital Technology	750	4
Healthcare and AI	Eshwar	2019	AI	950	9
Introduction to Data Science	Dravid	2014	Data Science	400	5

DSE (PUE)

```
INSERT INTO Library  VALUES('The Data Science Handbook', 'Darshan', 2019, 'Data science', 750.00, 12)
INSERT INTO Library  VALUES ('Introduction to Computer Programs', 'Harshavardhan', NULL, 'Computer Science', 700.00, 6)
INSERT INTO Library  VALUES ('Computer Science Text book Class 12', 'Reeta Sahoo', 2019, 'Textbook', 450.00, 4)
INSERT INTO Library  VALUES ('A book on AI', 'Sagar', 2016, 'AI', 200.00, 7)
INSERT INTO Library  VALUES ('Robots and Automation', 'Dushyanth', 2018, 'Digital Technology', 750.00, 9)
INSERT INTO Library  VALUES ('AI 2041', 'Chen & Lee', 2021, 'AI', 1000.00, 2)
INSERT INTO Library  VALUES ('Computer Hardware and Software', 'Chethan', 2000, 'Computer science', 500.00, 7)
INSERT INTO Library  VALUES ('The Future of Work', 'Dev Kumar', 2018, 'Digital Technology', 750.00, 4)
INSERT INTO Library  VALUES ('Healthcare and AI', 'Eshwar', 2019, 'AI', 950.00, 9)
INSERT INTO Library  VALUES ('Introduction to Data Science', 'Dravid', 2014, 'Data Science', 400.00, 5);
```

(i) List all the books

```
select * from library;
```

(ii) Calculate Amount by altering table by adding a new column ‘Amount’

(a) alter table library add(amount float(7,2));

(b) update library set amount = price * qty;

(iii) List the records of all those books price is between 400 and 900.

select price from library where price between 400 and 900;

(iv) List those records with no value in the attribute year .

select title, year from library where year is NULL;

(v) List the names of the authors whose name starts with letter ‘C’ or ‘D’.

select Author from library where author like 'C%' or author like 'D%';

(vi) List Title, year and category from the table library with category field has word ‘science’.

SELECT Title, Year, Category FROM Library WHERE Category LIKE '%science%';

(vii) List all those records whose year of publication is 2010 onwards with book price is less than Rs.750.

SELECT year, price FROM Library WHERE Year >= 2010 AND Price < 750;

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